



SynthVLM: Towards High-Quality and Efficient Synthesis of Image-Caption Datasets for Vision-Language Models

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Abstract

Vision-Language Models (VLMs) have recently emerged, demonstrating remarkable vision-understanding capabilities. However, training these models requires large-scale datasets, which brings challenges related to efficiency, effectiveness, and quality of web data. In this paper, we introduce SynthVLM, a new data synthesis and curation method for generating image-caption pairs. Unlike traditional methods, where captions are generated from images, SynthVLM utilizes advanced diffusion models and high-quality captions to synthesize and select images from text captions, thereby creating precisely aligned image-text pairs. We further introduce SynthVLM-100K, a high-quality dataset consisting of 100K curated and synthesized image-caption pairs. In both model and human evaluations, SynthVLM-100K outperforms traditional real-world datasets. Leveraging this dataset, we develop a new family of multimodal large language models (MLLMs), SynthVLM-7B and SynthVLM-13B, which achieve state-of-the-art (SOTA) performance on various vision question-answering (VQA) tasks. Notably, our models outperform LLaVA across most metrics with only 18% pretrain data. Furthermore, SynthVLM-7B and SynthVLM-13B attain SOTA performance on the MMLU benchmark, demonstrating that the high-quality SynthVLM-100K dataset preserves language abilities. Our dataset and the complete data generating and curating methods can be found in <https://github.com/stariver030515/SynthVLM>.

CCS Concepts

• Information systems → Information integration.



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Keywords

Synthetic Data, Data Curation, Vision Language Models

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1 Introduction

In recent years, with the rapid advancements in large language models (LLMs) [33, 38] and multimodal large language models (MLLMs) [43, 46], data management has become a crucial aspect of these technologies [5, 10, 18, 30, 32, 39]. At the same time, [2, 47–49] also demonstrates that data processing, selection, and management can significantly influence the performance of MLLMs.

Vision Language Models (VLMs), a subset of MLLMs, excel in tasks like image classification, understanding, and captioning [1, 6, 20, 21]. While most VLMs focus on improving model architectures to integrate multimodal data [1, 6, 20, 21, 26, 27], their reliance on web-sourced data poses a bottleneck: high-quality, precisely aligned image-text pairs are scarce. Noisy or misaligned data directly limits performance, as shown by [22, 41], who link enhanced VLM effectiveness to carefully curated datasets. Crucially, precise alignment between modalities underpins data quality, which is a challenge demanding greater focus than architectural refinements alone.

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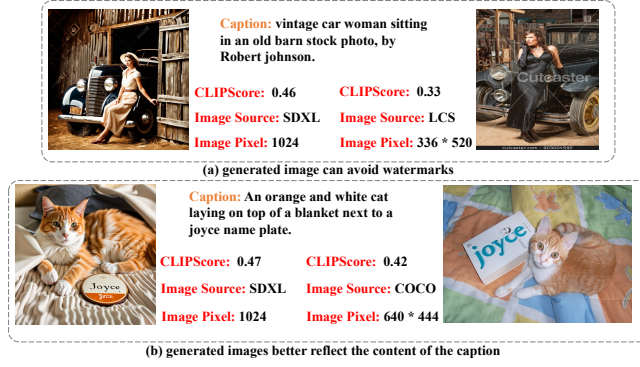


Figure 1: We compared SynthVLM-100K with LLaVA-558K. In (a), generated image can avoid content such as watermarks and advertisements. In (b), the generated images better reflect the content of the captions. Additionally, the resolution of the generated images is higher than real images.

To address the scarcity of high-quality web data, data synthesis strategies have increasingly been employed [3, 13, 31]. For instance, [31] utilized BLIP2 to generate numerous image captions, achieving SOTA results on DataComp. In the domain of VLMs, [3] used GPT-4 Vision to produce highly descriptive image captions, resulting in significant improvements in LLaVA. The integration of these generative models has opened new avenues for enhancing data quality and alignment, further boosting VLM performance.

Despite these notable advancements in VLMs, the challenge of multimodal data persists, particularly when paired image and text are required. Although synthetic captions have been generated for images, the absence of generated images means that the issue of limited high-quality web images remains unresolved. This limitation gives rise to three key challenges that still need to be addressed:

C1. Low Data Quality. Web-sourced images often contain artifacts such as blurriness and watermarks, compromising quality. Furthermore, caption generation approaches [31] leveraging BLIP2 tend to introduce logical inconsistencies and grammatical errors in text, which propagate through VLM training pipelines, ultimately weakening their linguistic reasoning capabilities.

C2. Poor Effectiveness. Existing datasets frequently lack the diversity and depth required to train VLMs effectively. Many web-scraped datasets consist of low-quality or irrelevant content that fails to capture the complexity of natural language and visual information. Furthermore, the limited scope of current datasets often results in models that struggle to generalize across various contexts.

C3. Low Efficiency. Methods that rely on manual captioning are both labor-intensive and resource-intensive. Automated solutions such as ShareGPT4V [3], which leverage GPT-4 for labeling, are costly and difficult to scale. Additionally, many strategies require the creation of large datasets, resulting in significant data redundancy.

To address these challenges, we introduced a new data generation pipeline, SynthVLM. First, we implemented a quality selection process to filter high-quality caption data. Next, we employed diffusion models to generate images based on these captions. For quality

assurance, we proposed a new method for evaluating and selecting image-caption pairs by combining CLIPScore [17] and SSIM [42], focusing on both image-text alignment and image quality. Our data generation approach achieved superior alignment between images and captions compared to existing methods. With 100K curated synthetic data, we attained SOTA results across multiple benchmarks, utilizing only 18% of the official LLaVA-558K dataset size. Overall, our contributions are as follows:

- **Innovative Data Synthesis Framework.** We introduce SynthVLM, a novel two-stage pipeline for generating synthetic image-caption pairs, and curate SynthVLM-100K—a large-scale, high-quality dataset produced through this framework. Compared to conventional approaches, SynthVLM achieves superior efficiency without compromising quality.
- **Superior Synthetic Data Quality.** SynthVLM-100K outperforms baseline datasets in both model and human evaluations. It achieves superior CLIPScore and SSIMScore, while also earning higher human evaluation ratings. Further validation via GPT-4 and Intern-VL2 assessments corroborates its enhanced quality for training vision-language models.
- **State-of-the-Art Vision-Language Modeling.** When pre-trained exclusively on SynthVLM-100K, our models surpass baseline methods relying on LLaVA-558K, achieving top-tier performance on vision-language understanding benchmarks. SynthVLM-13B also delivers exceptional performance on the MMLU benchmark, highlighting its multimodal versatility.

2 Related Work

2.1 Data Quality and Selection

High-quality data can significantly enhance the performance of models [29]. Ensuring high data quality becomes more challenging because it requires more resources for data cleaning, selection and annotation [2]. LLMs-based methods were commonly used in data selection [2]. For instance, [8] leverages DeBERTa [15] for scoring, retaining high-quality data, and combining it with the k-center greedy algorithm to select diverse data. [4] scores the accuracy of data using ChatGPT to pick out high-quality data. [44] use GPT-4 to rewrite data to increase their complexity and then streamline it by reducing its variety and improving its quality.

2.2 Data Generation

Recent advancements in generating synthetic data and improving the performance of LLMs have shown promising results across various domains. A key component in generating high-quality synthetic datasets is precise alignment. [9] introduce REALIGN, a method that enhances the quality of instruction data by reformatting responses to better align with pre-established criteria and evidence, thereby improving LLMs' alignment with human values while minimizing human annotation and model hallucinations. [23] build a high-quality instruction-following language model by automatically labeling human-written text with corresponding instructions and demonstrating highly effective self-alignment.

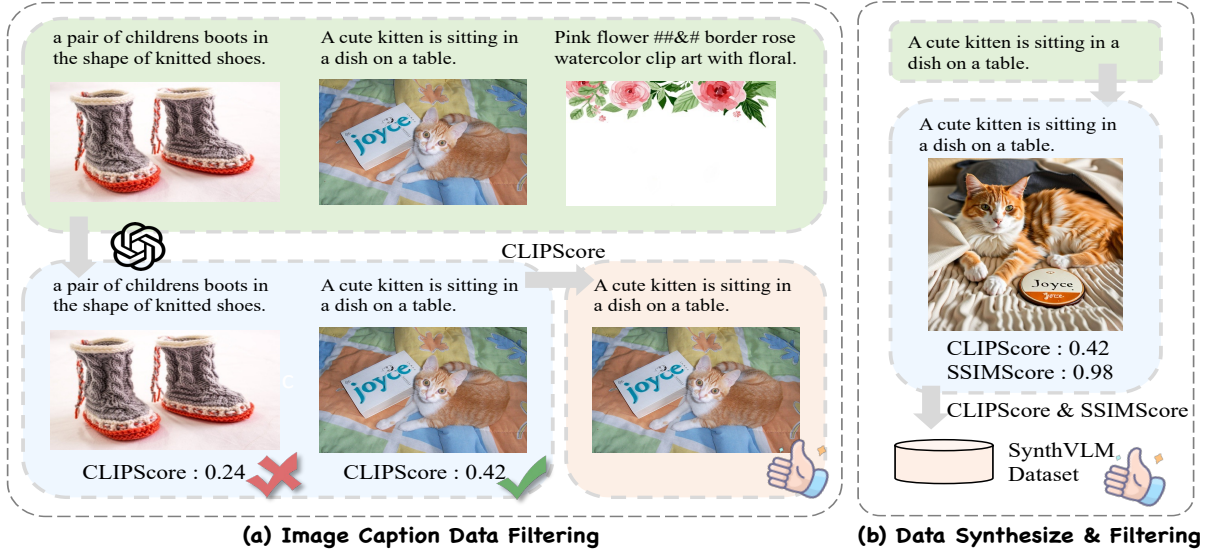


Figure 2: The pipeline of the SynthVLM data synthesis method is as follows: First, we filter high-quality image-caption pairs. Next, we synthesize high-quality data and subsequently filter them based on CLIPScore.

Table 1: LCS abbreviates the LAION, CC, and SBU datasets. SynthVLM uses captions to generate images, while others use images to generate captions or manual labeling.

Name	Image Source	Caption Source	Sample
COCO-Caption [25]	COCO	Human	118K
BLIP-LCS [26]	LCS	BLIP	558K
ShareGPT4V [3]	LCS, COCO, etc	GPT4-Vision	100K
ShareGPT4V-PT [3]	LCS, COCO, etc	Share-Captioner	1246K
SynthVLM	Diffusion	LCS, COCO, BLIP2-DataComp, etc	1000K

3 Method

3.1 Step1: Synthetic Dataset Construction

In this section, we introduce the image generation pipeline. First, we construct a large pool of captions. We then select the best captions from the pool for image-text generation. Utilizing these high-quality captions, we employ diffusion models to generate the images.

Data Source. To ensure the diversity of the captions, we combined human-generated and model-generated captions. As shown in Table 1. The human-generated captions were primarily sourced from LAION, CC, and SBU, while the model-generated captions were created using the method described in [31], which utilizes BLIP2 to regenerate captions for images in the DataComp dataset [12].

Caption Curation. To maintain dataset quality, we first removed low-quality captions, such as advertisements, overly repetitive descriptions, and captions with significant grammatical errors. The filtering process was performed with ChatGPT, combined with statistical indicators such as N-grams and Perplexity, ensuring that only high-quality, informative captions were used for training. For the remaining captions, we calculated the CLIPScore[17] for these captions and their corresponding raw images. CLIPScore is a metric that measures the cosine similarity between images and their

corresponding captions. The formula for calculating CLIPScore is as follows:

$$CLIPScore(I, C) = \frac{CLIP(I) \cdot CLIP(C)}{\|CLIP(I)\| \cdot \|CLIP(C)\|} \quad (1)$$

where I represents the image, C represents the caption, and $CLIP(I)$ and $CLIP(C)$ denote the image and text feature vectors extracted by the CLIP model. The dot product of the vectors is denoted by $\cdot$, and $\|\cdot\|$ denotes the norm of the vectors.

We selected the top 40% of image-caption pairs with the highest CLIPScores. These selected captions were included in the candidate caption set. Ultimately, we curated a dataset of 1M captions for data generation. By using only captions, our method significantly reduces storage overhead and processing time. The caption curation pipeline is summarized in Figure 2(a).

Image Generation. After filtering 1M high-quality captions, we employed Stable Diffusion XL (SDXL) [35], a SOTA model capable of efficiently generating high-quality, high-resolution images. SynthVLM produces images at a resolution of 1024x1024, effectively addressing the low-resolution issues present in existing datasets. This improvement greatly enhances the quality and utility of the training data across various image generation and recognition tasks.

3.2 Step2: Synthetic Data Selection

In this section, we introduce a novel algorithm for quality control of generated datasets. To better ensure the alignment between images and their corresponding text descriptions, we continue to use CLIPScore. For a given image I and its corresponding text C , we first calculate the image-text CLIPScore, $CLIPScore(I, C)$, to assess their alignment. Additionally, the generated images have a resolution of 1024×1024 , which will be resized to 336×336 for compatibility with CLIP [36]. To account for the potential loss in quality due to this resizing process, we introduce the Structural

Table 2: Comparison of SynthVLM and LLaVA using the same model structure. We can see SynthVLM outperforms LLaVA on all the evaluation benchmarks.

Models	LLM	SQA	SQA ^I	MMVet	VizWiz	VQAv2	GQA	MMB	MME ^P	MME ^C	PoPE	MMLU
LLaVA-7B	Vicuna-1.5-7B	69.3	67.3	30.5	49.9	78.7	62.5	65.3	1484.8	315.6	86.0	36.3
SynthVLM-7B	Vicuna-1.5-7B	70.4	68.9	32.2	49.3	79.4	63.1	66.8	1518.5	345.7	87.0	41.2
LLaVA-13B	Vicuna-1.5-13B	74.2	71.0	35.0	53.6	80.0	63.0	67.7	1531.3	294.5	86.9	52.4
SynthVLM-13B	Vicuna-1.5-13B	74.9	72.5	35.0	55.9	80.0	63.5	68.3	1573.0	316.1	88.4	54.6

Table 3: Result comparison of MMLU shows that with the synthetic 100k data, our SynthVLM outperforms LLaVA in pure language tasks. This demonstrates the effectiveness of the synthetic data in modality alignment.

Models	LLM	SQA	MMLU				
			Avg	STEM	Humanities	Social Sciences	Other
LLaVA-7B	Vicuna-1.5-7B	69.3	36.3	28.6	33.4	39.5	44.5
SynthVLM-7B	Vicuna-1.5-7B	70.4	41.2	31.7	37.4	47.0	50.2
LLaVA-13B	Vicuna-1.5-13B	74.2	52.4	41.9	45.8	62.9	61.8
SynthVLM-13B	Vicuna-1.5-13B	74.9	54.6	45.0	49.3	64.0	62.2

Table 4: We compared the average CLIPScore, SSIMScore, and their weighted score across our synthetic dataset, ShareGPT4V, COCO-Caption, and BLIP-LCS. The results demonstrate that SynthVLM achieves the highest alignment and image quality.

Datasets	# Samples	CLIPScore	SSIMScore	Weighted_Score
COCO-Caption	118K	0.31	0.73	0.67
BLIP-LCS	558K	0.32	0.75	0.70
ShareGPT4V	100K	0.32	0.79	0.71
Synth Dataset	1000K	0.34	0.78	0.73
Curated-Synth Dataset	100K	0.36	0.86	0.79

Similarity Index Measure (SSIM), a metric for image quality control. This will help us better ensure the image quality is preserved after resizing. The SSIM formula is defined as follows:

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

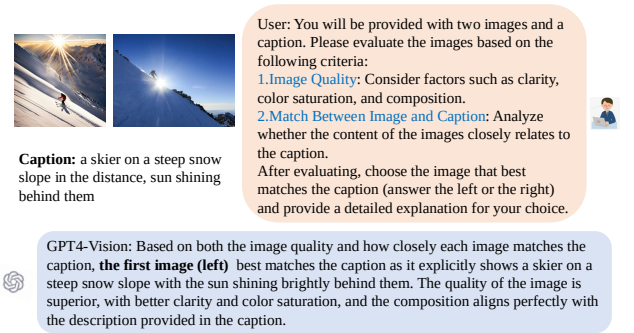
Where x and y are the two images being compared, μ_x and μ_y are the mean pixel intensities of the images, σ_x^2 and σ_y^2 are the variances of the images, σ_{xy} is the covariance of the images, C_1 and C_2 are constants to stabilize the division with weak denominator values.

For a given image I with a resolution of 1024×1024 , we first resize it to 336×336 , then interpolate to restore it back to 1024×1024 . The SSIM value is then computed between the resized image and the original image to quantify the loss introduced by the resizing process. Let $I_{resized}$ denote the resized image, where:

$$I_{resized} = \text{resize}(\text{resize}(I, (336, 336)), (1024, 1024))$$

The SSIMScore is computed as:

$$SSIMScore = SSIM(I, I_{resized})$$

**Figure 3: Our process and prompt design for match assessment using GPT4V. We consider various aspects, including the quality of the image and the match between the image and the caption. Based on this process, we compare SynthVLM with existing datasets from the model’s perspective.**

Finally, to combine the CLIPScore and SSIMScore, we apply a weighted sum, with SSIMScore weighted by a factor λ . The overall score S is given by:

$$\text{Weighted_Score} = \text{CLIPScore}(I, C) + \lambda \cdot \text{SSIMScore}(I, I_{resized})$$

In practice, we set $\lambda = 0.5$ to balance the contribution of CLIPScore and SSIMScore in the final score.

As shown in Figure 2(b), we initially computed CLIPScores and SSIMScore for the 1M synthetic image-caption pairs. We then selected the top 100K pairs that demonstrated the highest scores, indicating the most accurate and meaningful matches between images and captions. By curating this subset, we constructed a high-quality, highly aligned synthetic dataset.

Table 5: The results on the MME benchmark demonstrate that using generated data can still maintain a leading performance in real-world problems, further expanding the application scope of SynthVLM.

Models	LLM	MME Cognition			MME Perception					
		Reasoning	Translation	Code	Posters	Celebrity	Scene	Landmark	Artwork	OCR
LLaVA1.5-7B	Vicuna-1.5-7B	126.4	57.5	62.5	148.3	132.1	143.0	141.8	123.8	100.0
SynthVLM-7B	Vicuna-1.5-7B	143.6	70.0	57.5	157.1	124.2	157.8	156.5	136.2	102.5
LLaVA-13B	Vicuna-1.5-13B	119.3	50.0	62.5	155.4	127.4	158.5	165.2	129.3	110.0
SynthVLM-13B	Vicuna-1.5-13B	130.0	65.0	62.5	160.4	138.2	162.0	158.8	140.3	104.5

Table 6: We employed GPT4-Vision and InternVL to vote on the match between each caption and its corresponding generated image and raw image. The results demonstrate that the generated images align more closely with the captions.

Sample	Model	Synthetic Image win	Web Image win
1K	GPT4-Vision	633	367
1K	InternVL2	692	308
1K	Human	758	242

3.3 High Quality Synthetic Dataset

In this section, we compare commonly used image-caption datasets with the SynthVLM-100K dataset. The synthetic data offers high image quality, excellent image-text alignment and superior model ratings.

High Image Quality. As illustrated in Figure 1, SynthVLM markedly improves image quality by generating images at a resolution of 1024x1024 pixels. This high resolution addresses the prevalent issue of suboptimal image quality in existing datasets, providing high-quality image-caption pairs that are invaluable for training VLMs. Additionally, SynthVLM-100K effectively eliminates artifacts such as watermarks and advertisements.

Excellent Image-Text Alignment. As shown in Table 4, the SynthVLM-100K dataset exhibits a higher CLIPScore and SSIM-Score compared to existing high-quality web datasets. By selecting curated image-text pairs of superior quality, SynthVLM-100k surpasses datasets such as COCO-Caption, BLIP-LCS, and ShareGPT4V. This demonstrates the outstanding alignment of our dataset.

Excellent Model Rating. Since our data will be used for VLMs training, we use VLMs to evaluate the data quality. We selected 1K image-caption pairs and submitted the caption along with the synthetic image and the original image. We used human annotators, GPT-4 Vision[34] and Intern-VL2[40] as the judge model and requested it to select the pair that exhibited higher alignment. The specific prompt used for this evaluation is illustrated in Figure 3. The results, presented in Table 6, demonstrate that images generated have better alignment with the caption.

Through these two stages, we successfully developed SynthVLM-100K data. Our method is efficient, utilizing only 100K pre-training data. Additionally, SynthVLM provides a new paradigm for effective alignment between modalities in Vision Language Models using pure synthetic data.

Table 7: Comparison of data utilization for generating image-caption pairs. This indicates that our SynthVLM have superior efficiency compared to other methods.

Methods	SynthVLM	LLaVA	w/o selection
Dataset Number (k)	100	558	1000
Data Usage	33MB	27GB	310MB

4 Experiments

In this section, we utilize the image data synthesis system for various tasks. We then aim to answer the following questions to verify the effectiveness and efficiency of our proposed SynthVLM: **Q1:** Can our SynthVLM-7B and SynthVLM-13B achieve SOTA performance compared to previous SOTA methods? **Q2:** How does the efficiency of our SynthVLM compare to previous methods? **Q3:** Would using generated data in full impact the application of the model in real-world scenarios? **Q4:** Do we need the generate module and the quality selection module to enhance model performance?

4.1 Experimental Settings

Models. We use the LLaVA 1.5[26] model to validate the effectiveness of our dataset. For the Vision Encoder, we select CLIP 336px, and for the LLM, we use Vicuna 7B and Vicuna 13B[7]. Our training parameters are consistent with LLaVA, and we adopt a two-stage training approach. In the pre-training stage, we train the projector to align the image and text modalities. The SynthVLM-100K dataset described in Section 3.3 is used for this purpose. In the SFT stage, we further train the projector alongside the LLM to improve visual understanding capabilities.

Datasets. For LLaVA-BaseLine, we use the LLaVA 558k dataset for pre-training and the LLaVA 665k dataset for SFT. For SynthVLM-7B and SynthVLM-13B, we use the SynthVLM-100K dataset for pre-training and the LLaVA 665k dataset for SFT.

Benchmarks. We select benchmarks for both visual understanding and language understanding. For visual understanding, we choose SQA^I[28], MMVet[45], VizWiz[14], VQAv2[37], GQA[19], MME[11], and PoPE[24] for a comprehensive evaluation. For language benchmarks, we select MMLU[16] and SQA[28] to assess language understanding abilities.

Settings. We primarily adhered to the hyperparameters specified in the official repository for the LLaVA model evaluations. All

Table 8: Ablation study of visual understanding ability and pure language ability. The results demonstrate that removing either the data generation or data selection module results in a performance drop.

Models	LLM	SQA	SQA ^I	MMVet	VizWiz	VQAv2	GQA	MMB	MME ^P	MME ^C	PoPE	MMLU
SynthVLM-7B	Vicuna-1.5-7B	70.4	68.9	32.2	49.3	79.4	63.1	66.8	1518.5	345.7	87.0	41.2
w/o generation	Vicuna-1.5-7B	69.3↓	67.0↓	31.2↓	46.8↓	79.3↓	62.9↓	66.2↓	1488.8↓	327.5↓	86.2↓	39.1↓
w/o selection	Vicuna-1.5-7B	69.9↓	67.7↓	30.2↓	50.2	79.1↓	62.2↓	63.5↓	1421.7↓	301.8↓	87.3	40.6↓
SynthVLM-13B	Vicuna-1.5-13B	74.9	72.5	35.0	55.9	80.0	63.5	68.3	1573.0	316.1	88.4	54.6
w/o generation	Vicuna-1.5-13B	73.6↓	71.4↓	33.0↓	53.6↓	80.0	63.4↓	67.5↓	1514.3↓	295.7↓	88.2↓	53.6↓
w/o selection	Vicuna-1.5-13B	74.1↓	70.5↓	35.6	53.2↓	79.7↓	63.1↓	67.5↓	1512.7↓	303.2↓	86.9↓	53.0↓

Table 9: Ablation study of modality alignment. The results demonstrate that removing either the data generation or data selection module results in a performance drop.

Models	SQA	MMLU				
		Avg	STEM	Humanities	Social Sciences	Other
SynthVLM-7B	70.4	41.2	31.7	37.4	47.0	50.2
w/o generation	69.3↓	39.1↓	30.0↓	36.6↓	43.1↓	47.3↓
w/o selection	69.9↓	40.6↓	30.8↓	37.2↓	45.3↓	48.9↓
SynthVLM-13B	74.9	54.6	45.0	49.3	64.0	62.2
w/o generation	74.1↓	53.6↓	43.5↓	48.2↓	63.1↓	61.8↓
w/o selection	73.6↓	53.0↓	42.9↓	46.8↓	63.8↓	61.3↓

experiments were conducted on an 8x NVIDIA A100 GPU machine equipped with a 120-core CPU and 960GB of memory.

4.2 Synthetic Data Achieves SOTA Performance

To address **Q1**, we trained the model described in experimental settings. From Table 2, it is evident that SynthVLM model outperforms the Baseline across all evaluation benchmarks on both 7B and 13B. SynthVLM model also excels in language benchmarks, demonstrating superior performance in SQA and MMLU, thus showcasing its comprehensive capabilities in both vision and language tasks.

These results demonstrate the strong alignment capability of our synthetic data. Additionally, this provides a new paradigm for effective visual understanding model modality alignment using generated data. During pre-training, it is common to train on all available data due to uncertainty about data selection. Here, we offer 100k high-quality synthetic data as a benchmark for selecting aligned generated data efficiently.

4.3 Efficient Vision Language Alignment

To address **Q2**, we examine the computational resource usage during training and evaluate the data utilization efficiency for generating image-caption pairs.

As shown in Figure 7, by integrating a data selection module, our approach utilizes only 19% of the LLaVA data and 10% of the original synthetic data while achieving SOTA performance. This demonstrates that our data selection method can reduce computational usage by more than 80%.

4.4 Capability to Solve Real-World Problems

A critical issue when using generated data is whether the model loses its ability to solve real-world problems. To address **Q3**, we

utilized the MME Benchmark to assess the model’s performance in various real-world scenarios, such as artwork, celebrity, code reasoning, landmarks, and posters. The specific results are presented in Table 5. Using SynthVLM-100K, our model performs consistently well, matching or even surpassing the baseline across most real-world tasks. We attribute this success to the diversity of the captions and the generalization ability of the Diffusion model.

4.5 Ablation Study

To address **Q4**, We conducted an ablation study where we removed the data generation module and the data selection module separately to evaluate their individual contributions to the effectiveness of our data generation pipeline. In this section, we controlled the number of samples in each experimental group to be 100K.

Excluding Data Generation Module. The exclusion of the data generation module significantly impacts the model’s performance, as illustrated in Tables 8 and 9, labeled as "w/o generation". The variant without this module demonstrates lower accuracy, emphasizing the crucial role of the data generation process in sustaining the high performance of the SynthVLM model. This also underscores SynthVLM’s potential in constructing highly aligned datasets.

Excluding Data Selection Module. The absence of the data selection module similarly leads to a noticeable decline in performance, indicated as "w/o selection" in Tables 8 and 9. Given the inherent randomness of diffusion models, which inevitably generate some low-quality images, the data selection module is crucial for removing these subpar elements.

Overall, the ablation study highlights the critical role of data generation and data selection in SynthVLM, providing valuable insights into the contributions of each module.

5 Conclusion

We propose SynthVLM, a novel pipeline for generating high-quality pre-training data for VLMs. Unlike existing datasets, our synthetic images are free from watermarks and advertisements, leading to cleaner visual inputs. The generated data demonstrate superior alignment and visual fidelity. Notably, the SynthVLM model—trained with only 18% synthetic data—outperforms the LLaVA baseline trained on the full dataset. It not only achieves state-of-the-art alignment performance but also preserves the language understanding capabilities of VLMs. Ablation studies further confirm the effectiveness of both the image generation and data selection modules, highlighting the practical value of our data synthesis approach.

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